Data-driven SPS/DPS separation

From the ATLAS 8 TeV measurement

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July 31, 2026

Can data replace an imperfect MC template?

MC-template limitation

- SPS modeling is not fully reliable
- Template choices propagate to the fraction
- A complementary data constraint is desirable

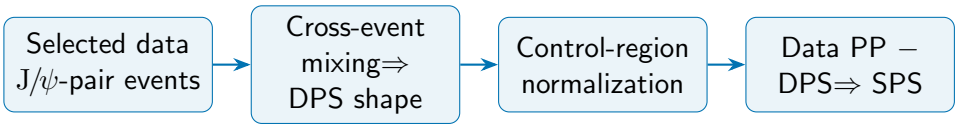
Exploratory goal

- Construct DPS from mixed Data candidates
- Normalize in a DPS-enhanced region
- Derive SPS by subtraction

Explore whether a data-driven construction can replace the MC method when the simulation is not fully reliable.

ATLAS Collaboration, Eur. Phys. J. C 77 (2017) 76, arXiv:1612.02950

Data-driven construction method from ATLAS



DPS assumption

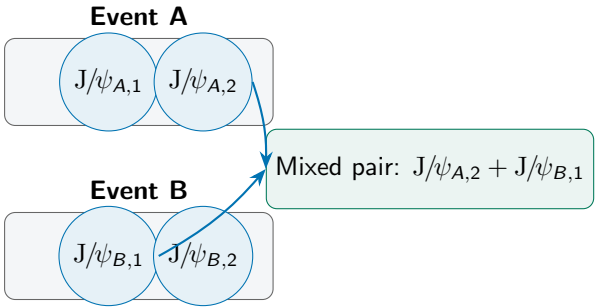
$$P_{\mathrm{DPS}}(J/\psi_1, J/\psi_2) \simeq P(J/\psi_1)P(J/\psi_2).$$

Data determine

$$N^i_{\mathrm{SPS}} = N^i_{\mathrm{PP}} - \alpha N^i_{\mathrm{mix}}.$$

ATLAS Collaboration, Eur. Phys. J. C 77 (2017) 76, arXiv:1612.02950

Event mixing starts from selected candidates, not a pure PP sample



- Start from **2492 selected pair candidates**: prompt, nonprompt, and combinatorial components may all be present
- Select two **different source events**
- Take one reconstructed J/ψ from each event
- Recompute the mixed-pair selection and all pair-level observables
- Remove within-event SPS correlations

Prompt–prompt yields are extracted only in the later four-dimensional fits.

Mixing preserves inclusive single-J/ψ marginals

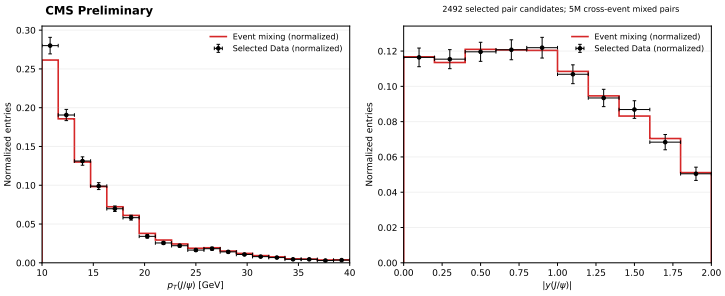


Figure: single-J/ψ marginals

Meaning: after cross-event recombination and reapplying pair selection, the inclusive one-dimensional kinematics remain stable.

Why this check? Mixed pairs are reselected and their pair observables change; stable single-J/ψ marginals show that mixing has not visibly distorted the source-candidate kinematics.

CDF distances: (p_T)=0.017 (—y—)=0.00461

Prompt–prompt yields in the $|\Delta y|$–$|\Delta\phi|$ plane

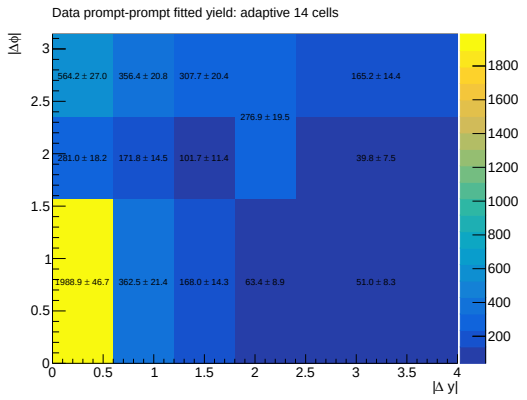


Figure: Data PP fitted yields

Meaning: each cell shows the prompt–prompt yield from a four-dimensional mass–lifetime fit.

For each 2D cell

- Simultaneous 4D fit in $(m_1, m_2, c\tau_1, c\tau_2)$
- Extract PP data yield N^i_{PP}
- Use identical cells for data and mixed DPS

Why adaptive cells?

Merge sparse regions until every cell supports a stable fit, while retaining the control-region boundaries and correlation structure.

Normalization in a DPS-dominated control region

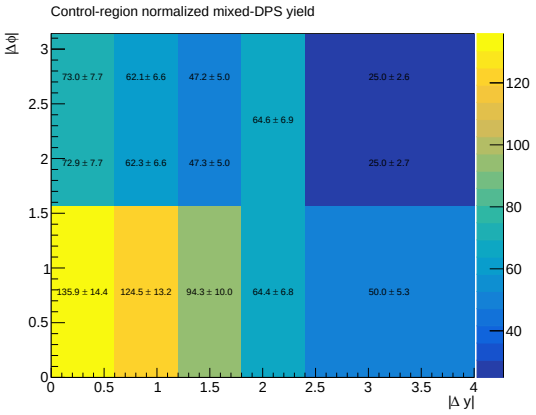


Figure: normalized mixed-DPS yields

Meaning: the control region fixes the mixed-DPS normalization in every cell.

Control region

$$|\Delta y| \geq 1.8, \qquad |\Delta\phi| \leq \pi/2.$$

Normalization

$$\alpha = \frac{\sum_{i\in \mathrm{CR}} N^i_{\mathrm{PP}}}{\sum_{i\in \mathrm{CR}} N^i_{\mathrm{mix}}}, \qquad N^i_{\mathrm{DPS}} = \alpha N^i_{\mathrm{mix}}.$$

- Large rapidity separation enriches DPS
- Small azimuthal separation suppresses back-to-back SPS

ATLAS Collaboration, Eur. Phys. J. C 77 (2017) 76, arXiv:1612.02950

SPS weight by subtraction

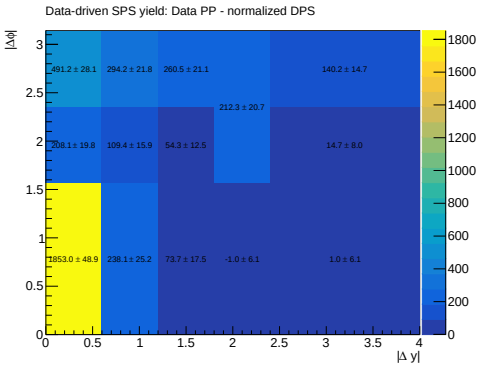


Figure: data-driven SPS yields

Meaning: subtract normalized DPS from Data PP cell by cell.

$$N^i_{\mathrm{SPS}} = N^i_{\mathrm{PP}} - N^i_{\mathrm{DPS}}.$$

Why can one control-region cell be negative?

Data and scaled mixed yields fluctuate independently. Their estimator difference is not constrained cell by cell.

- Only one negative SPS cell
- Compatible with zero at -0.165σ
- Signed value avoids clipping bias
- Global integrals remain physical

SPS & DPS weights

Figure: SPS component weights

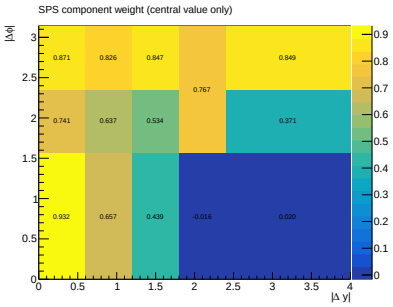
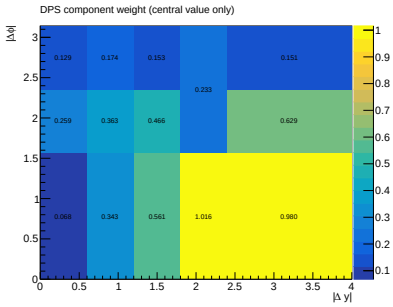


Figure: DPS component weights



Central-value construction only: full 14-cell weight covariance is not available.

$$w^i_{\mathrm{SPS}} = \frac{N^i_{\mathrm{SPS}}}{N^i_{\mathrm{PP}}}, \quad w^i_{\mathrm{DPS}} = \frac{N^i_{\mathrm{DPS}}}{N^i_{\mathrm{PP}}}, \quad w^i_{\mathrm{SPS}} + w^i_{\mathrm{DPS}} = 1.$$

Primary fraction: integral of the two-dimensional decomposition

$$f_{\mathrm{SPS}}^{\mathrm{primary}} = \frac{\sum_{i=1}^{14} N_{\mathrm{SPS}}^{i}}{\sum_{i=1}^{14} N_{\mathrm{PP}}^{i}}, \qquad f_{\mathrm{DPS}}^{\mathrm{primary}} = 1 - f_{\mathrm{SPS}}^{\mathrm{primary}}.$$

Component	Primary fraction
SPS	**0.806350**
DPS	**0.193650**

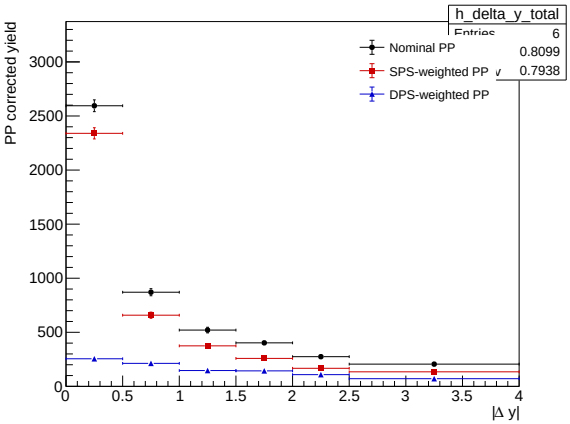
Definition

Sum the subtracted SPS and normalized DPS yields over all 14 cells, then divide by the total fitted PP yield.

- One integrated estimator
- No averaging over cells
- Same decomposition as the weight maps

"Primary" denotes the direct 2D integral before the 1D component refits.

From 2D weights to one-dimensional component yields

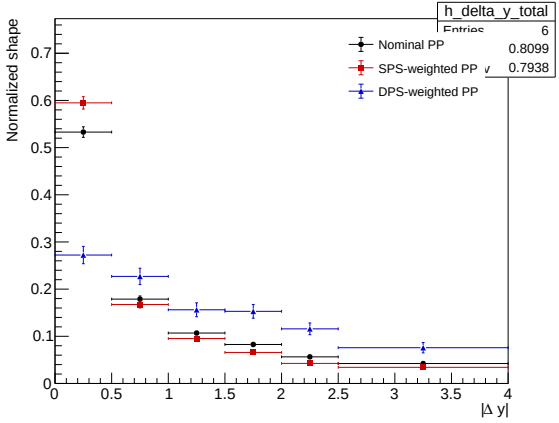


For every target bin

1. Assign its parent 2D-cell weight
2. Build SPS- and DPS-weighted data
3. Repeat the mass–lifetime fit
4. Extract component PP yields

Targets: $m(JJ)$, $|\Delta y|$, $|\Delta\phi|$, $|y(JJ)|$.

The reported fraction uses the $|\Delta y|$ distribution



Estimator	f_{SPS}
2D primary	0.806350
$\lvert\Delta y\rvert$	**0.807527**

$$\Delta f_{\mathrm{SPS}} = 0.001177 \simeq 0.39\sigma.$$

Interpretation

The four target variables share one 2D decomposition; they are consistency views, not independent measurements.

PP-corrected component yields: $|\Delta\phi|$

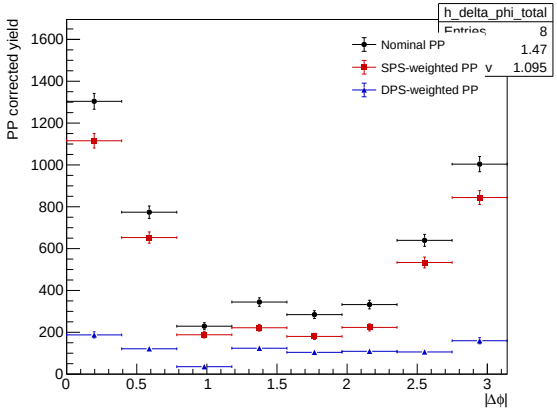


Figure: fitted PP yields

- Apply the same 2D SPS/DPS weights
- Refit PP yield in eight $|\Delta\phi|$ bins
- Vertical axis is PP-corrected yield, not a unit-area shape

PP-corrected component yields: $m(JJ)$

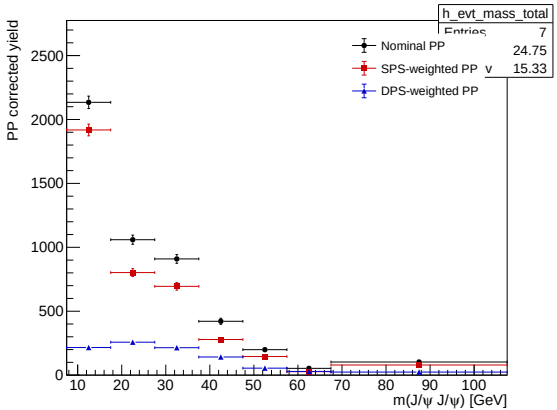


Figure: fitted PP yields

- Apply the same 2D SPS/DPS weights
- Refit PP yield in seven $m(JJ)$ bins
- Vertical axis is PP-corrected yield, not a unit-area shape

PP-corrected component yields: $|y(JJ)|$

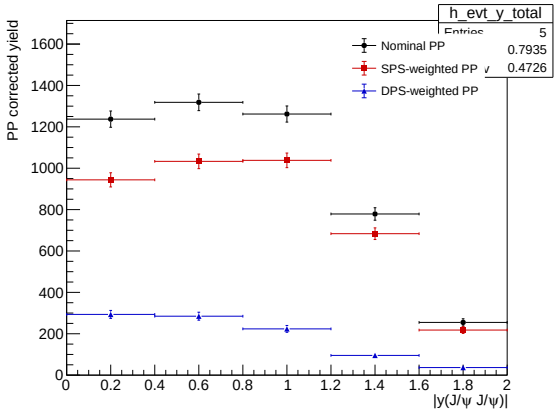
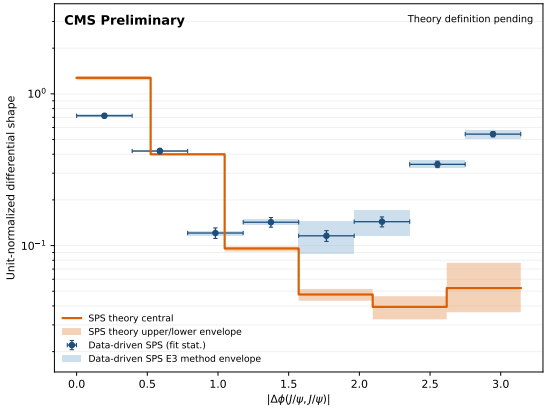


Figure: fitted PP yields

- Apply the same 2D SPS/DPS weights
- Refit PP yield in five $|y(JJ)|$ bins
- Vertical axis is PP-corrected yield, not a unit-area shape

Measured σ_{SPS} vs. theory prediction: $|\Delta\phi|$

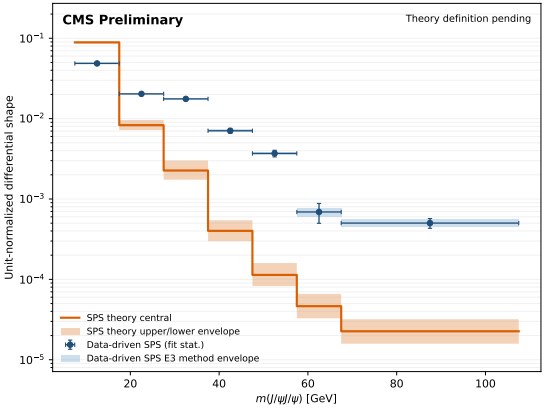


Working assumptions

- Theory: six equal-width bins on $[0, \pi]$
- Data: eight bins; no rebinning
- All theory variations and Data normalized to unit area
- Fiducial, units, and envelope definitions remain pending

Qualitative shape comparison only; no data/theory ratio.

Measured σ_{SPS} vs. theory prediction: $m(JJ)$

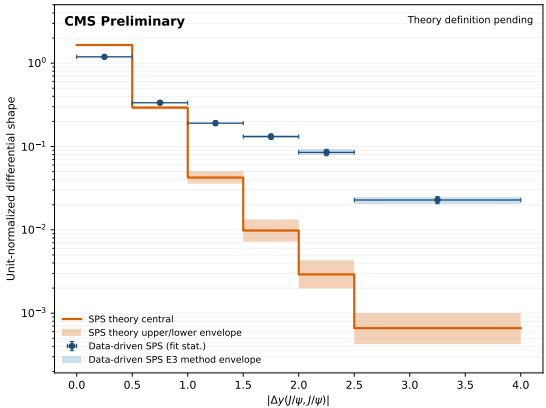


Working assumptions

- Theory: seven entries mapped to seven Data bins
- Identical theory and Data bin edges
- All theory variations and Data normalized to unit area
- Fiducial, density, and envelope definitions remain unconfirmed

Qualitative shape comparison only; no data/theory ratio.

Measured σ_{SPS} vs. theory prediction: $|\Delta y|$

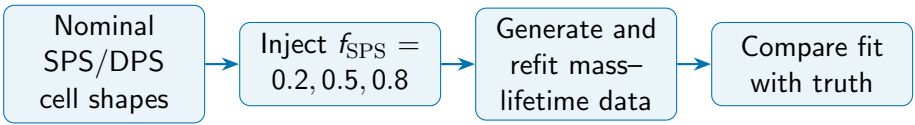


Working assumptions

- Original theory: eight bins of width 0.5
- Last three theory bins merged into 2.5–4.0
- Width-weighted merging preserves the theory integral
- All theory variations and Data normalized to unit area

Theory definitions remain pending; no data/theory ratio.

Parametric implementation closure with nominal shapes



Thirty ensembles per truth point

- Same nominal shapes and fit model
- Different random seed
- Different event counts and observables
- Initial bias/pull/coverage diagnostic

Scope

This is a full-chain implementation closure. It does not validate the event-mixing shape or the control-region purity.

Closure definitions and results

$f_{\mathrm{SPS}}^{\mathrm{true}}$	Complete	Bias	Pull width	1σ coverage
0.2	9/30	-0.00215	0.844	—
0.5	30/30	$+0.00095$	0.919	0.767
0.8	30/30	$+0.00289$	1.191	0.633

Definitions

$$\mathrm{bias} = \langle f_{\mathrm{fit}} \rangle - f_{\mathrm{true}}, \qquad \mathrm{pull} = \frac{f_{\mathrm{fit}} - f_{\mathrm{true}}}{\sigma_{\mathrm{fit}}}.$$

Coverage is the fraction with truth inside $f_{\mathrm{fit}} \pm \sigma_{\mathrm{fit}}$.

Completed ensembles

At 0.5 and 0.8, all chains complete with small bias at 30-ensemble precision.

Failed stress test

At 0.2, negative SPS components stop the chain after 9/30; no coverage claim is made.

Tested variations and SPS-leakage closure

Category	Nominal	Alternative
CR $\lvert\Delta\phi\rvert$	$\leq \pi/2$	$\leq 7\pi/18,\ 11\pi/18$
CR $\lvert\Delta y\rvert$	≥ 1.8	≥ 2.4
2D binning	adaptive 14 cells	coarse 10 cells
Mixing	seed 20260728	seed 20260731
SPS leakage	zero	MC estimate 0.0005621

What does the leakage closure test?

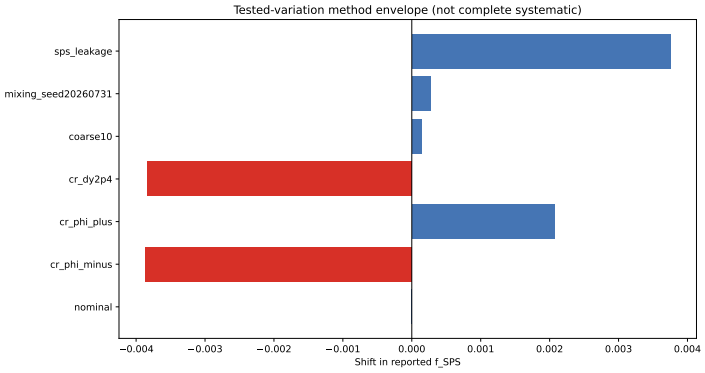
Test the pure-DPS control-region assumption by subtracting the MC-estimated SPS leakage before normalizing:

$$\alpha_{\mathrm{leak}} = \frac{N_{\mathrm{PP}}^{\mathrm{CR}} - f_{\mathrm{leak}} N_{\mathrm{PP}}^{\mathrm{all}}}{N_{\mathrm{mix}}^{\mathrm{CR}} - f_{\mathrm{leak}} N_{\mathrm{mix}}^{\mathrm{all}}}, \quad f_{\mathrm{leak}} = 0.0005621.$$

Propagation

Rebuild templates and weights, then rerun all four 1D component fits. The shift tests SPS-contamination sensitivity; it does not measure CR purity.

Tested-variation method envelope



Largest signed shifts

$$\Delta f_{\mathrm{SPS}}^{\mathrm{method}} = \begin{matrix} +0.003750 \\ -0.003867 \end{matrix}.$$

- Up: SPS leakage variation
- Down: narrower CR-phi
- Scan includes all six tested alternatives

Scope

This is the tested-variation method envelope, not a complete systematic uncertainty.

What the present envelope does — and does not — test

Evaluated or partially tested

- CR boundary dependence
- Adaptive14 versus coarse10
- Mixing resampling seed
- One nominal SPS leakage estimate
- Same-model implementation closure

Not evaluated end to end

- Trigger and muon reconstruction
- Acceptance/efficiency variations
- Alternative mass and PP lifetime models
- Differential PP-weight bias
- Pile-up
- Physical alternative to event mixing

Event mixing as DPS and CR purity remain only partially validated assumptions.

Shape envelope: largest tested bin-by-bin change

$$E_i = \max_{v} \frac{|p_i^{(v)} - p_i^{\mathrm{nom}}|}{p_i^{\mathrm{nom}}}.$$

For each normalized component bin, scan all tested variations and retain the largest absolute relative shift.

Variable	SPS	DPS
$\lvert\Delta y\rvert$	8.9%	18.9%
$\lvert\Delta\phi\rvert$	24.2%	41.8%
$m(JJ)$	11.9%	36.0%
$\lvert y(JJ)\rvert$	1.95%	11.4%

How it is used

The envelope supplies a tested-method band for differential SPS/DPS shapes. It is not added again as statistical uncertainty.

Interpretation

Largest relative changes occur mainly in low-population bins; they need not produce a large integrated-fraction shift.

From f_{DPS} to σ_{DPS} and σ_{eff}

DPS cross section

$$\sigma_{\mathrm{DPS}} = f_{\mathrm{DPS}}\,\sigma_{JJ}.$$

$$f_{\mathrm{DPS}} = 0.192473 \pm 0.018829\ (\mathrm{stat.})\ {}^{+0.003867}_{-0.003750}\ (\mathrm{tested\ method}).$$

$$\sigma_{JJ} = 37.2 \pm 0.6\ (\mathrm{stat.}) \pm 2.5\ (\mathrm{syst.})\ \mathrm{pb}.$$

Effective cross section

$$\sigma_{\mathrm{eff}} = \frac{1}{2}\frac{(\mathcal{B}\sigma_J)^2}{\mathcal{B}^2\sigma_{\mathrm{DPS}}}.$$

$$\mathcal{B}\sigma_J = 10.63 \pm 0.01\ (\mathrm{stat.}) \pm 0.09\ (\mathrm{bin\ syst.})\ \mathrm{nb}.$$

- Single- and pair-J/ψ fiducial regions are aligned
- Branching-fraction uncertainty cancels
- Add independent ATLAS luminosity uncertainty

Uncertainty propagation and contributions

Source	Treatment	Effect on σ_{DPS}	Effect on σ_{eff}
f_{DPS} statistical	statistical	9.78%	9.78%
CMS pair statistical	statistical	1.61%	1.61%
ATLAS single-J/ψ statistical	statistical	—	0.19%
f_{DPS} tested-method envelope	systematic	2.01%	1.95–2.01%
CMS pair non-BR systematic	systematic	6.68%	6.68%
ATLAS bin systematic	systematic	—	1.69%
ATLAS luminosity	systematic	—	2.26%
CMS pair BF contribution	removed	0.78% removed	BR cancels

Statistical combination

Independent listed statistical inputs are added in quadrature.

Systematic combination

Independent listed systematic inputs are added in quadrature; up/down swap for σ_{eff} because σ_{DPS} is in the denominator.

The fraction term is a tested-variation method envelope, not a complete systematic assessment.

Data-driven and current CMS AN nominal results

Data-driven results

$$f_{\mathrm{DPS}} = 0.1925 \pm 0.0188\ (\mathrm{stat.})\ ^{+0.0039}_{-0.0037}\ (\mathrm{tested\ method})$$

$$\sigma_{\mathrm{DPS}} = 7.160 \pm 0.710\ (\mathrm{stat.})\ ^{+0.499}_{-0.498}\ (\mathrm{syst.})\ \mathrm{pb}$$

$$\sigma_{\mathrm{eff}} = 2.2207 \pm 0.2202\ (\mathrm{stat.})^{+0.1667}_{-0.1670}\ (\mathrm{listed\ syst.})\ \mathrm{mb}$$

Current CMS AN nominal

$$f_{\mathrm{DPS}} = 0.34 \pm 0.10\ (\mathrm{stat.})\ ^{+0.02}_{-0.17}\ (\mathrm{syst.})$$

$$\sigma_{\mathrm{DPS}} = 12.6 \pm 3.7\ (\mathrm{stat.})\ ^{+1.1}_{-6.4}\ (\mathrm{syst.})\ \mathrm{pb}$$

$$\sigma_{\mathrm{eff}} = 1.2619 \pm 0.3706\ (\mathrm{stat.})\ ^{+0.6415}_{-0.1131}\ (\mathrm{syst.})\ \mathrm{mb}$$

Summary

1. **Data-driven construction** Cross-event mixing gives the DPS shape; control-region normalization and cell-by-cell subtraction give the SPS shape.
2. **Validated cross-check** Statistical propagation and same-model closure are complete; the tested variations define a method envelope, not a complete systematic assessment.
3. **Cross-check result**

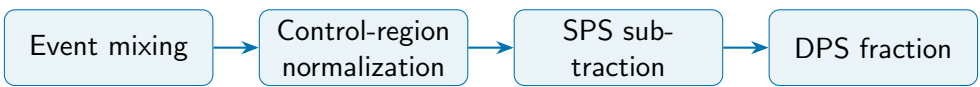
$$f_{\mathrm{DPS}} = 0.192473 \pm 0.018829\ (\mathrm{stat.})^{+0.003867}_{-0.003750}\ (\mathrm{tested\ method}),$$
$$\sigma_{\mathrm{eff}} = 2.2207 \pm 0.2202\ (\mathrm{stat.})^{+0.1667}_{-0.1670}\ (\mathrm{listed\ syst.})\ \mathrm{mb}.$$

A complete ATLAS-style data-driven SPS/DPS cross-check has been established for the selected CMS J/ψ-pair data.

Reference

Data-driven DPS method

ATLAS Collaboration, "Measurement of the prompt J/ψ pair production cross-section in pp collisions at $\sqrt{s} = 8$ TeV with the ATLAS detector, *Eur. Phys. J. C* **77** (2017) 76, arXiv:1612.02950.



A pseudoexperiment propagates one possible statistical fluctuation

One pseudoexperiment

Draw one correlated 14-cell Data-PP vector and one correlated 14-cell mixed-DPS vector from their covariance models.

1. Recompute CR normalization α
2. Recompute SPS/DPS cell yields and weights
3. Recompute primary and reported fractions

Repeat 50,000 times

The standard deviation of the resulting estimator distribution defines the statistical uncertainty.

- Data: diagonal fit covariance
- Mixing: source-event bootstrap covariance
- Fixed: nominal 1D response coefficients

The toys propagate the existing statistical information; they do not create new data.

Joint propagation: central values, uncertainties, and correlation

Estimator	Nominal f_{SPS}	Toy SD
2D primary	0.806350	0.020619
$\lvert\Delta y\rvert$ reported	**0.807527**	**0.018829**

Correlation

$$\rho = \frac{\mathrm{Cov}(f_{\mathrm{primary}}, f_{\mathrm{reported}})}{\sigma_{\mathrm{primary}}\sigma_{\mathrm{reported}}} = 0.992661.$$

Both estimators use the same Data and decomposition.

Toy bias

$$b = \langle f^{\mathrm{toy}}_{\mathrm{reported}} \rangle - f^{\mathrm{nominal}}_{\mathrm{reported}} = -0.001250.$$

This estimator-level bias is diagnosed further by the full parametric closure.